ISA104 | AI APPLICATION DESIGN

AI Applications: Voice, Speech & Creativity

Lecture 7 | AY 2026/27

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Learning Outcomes

By the end of this lecture, students should be able to:

01 **UNDERSTAND AI APPLICATIONS**

Identify common voice, speech and creative AI applications, and explain where AI contributes in their workflows.

02 **APPLY HUMAN-CENTRED DESIGN**

Apply human-centred design to your group AI Agent: identify users, needs, AI roles and human control.

Today's Journey

Start with people and tasks, then explore what AI can contribute

01

HUMAN-CENTRED DESIGN

Users · Tasks
Context · Needs

02

VOICE & SPEECH AI

How AI listens, understands and responds

03

CREATIVE AI

How AI helps people imagine, generate and make

04

YOUR GROUP AI AGENT

Apply what you learn to your group Agent

Anatomy of an AI Application

A simple way to understand the parts of an AI-powered experience

1 → 2 → 3 → 4 → 5

1	2	3	4	5
USER & TASK	**INPUT**	**AI CAPABILITY**	**OUTPUT / ACTION**	**HUMAN FEEDBACK**
Who needs help? What are they trying to do?	Voice · Text Image · Video	Recognise · Understand Generate · Decide	Information · Content Recommendation · Action	Accept · Edit Correct · Reject

AI is one part of the application — the user, task and feedback loop matter too.

Start with the Human: User, Task & Context

Before choosing an AI capability, understand the situation people are in

USER

Who will use it?

Roles, experience, goals and abilities.
Who else may be affected?

TASK

What are they trying to achieve?

Which steps take time, effort or lead to mistakes?
What does success look like?

CONTEXT

Where and when will it be used?

Device, noise, time pressure, shared spaces and connectivity.

Look for pain points: delays · repeated work · errors · uncertainty · accessibility barriers

Find out — don't assume: interview · observe · survey · analyse the task

Design framework: ISO 9241-210:2019 · Human-centred design for interactive systems

From Pain Point to Design

A human-centred cycle for turning user needs into an AI application

01

UNDERSTAND THE CONTEXT

Interview · Observe
Task analysis

→

02

SPECIFY USER NEEDS

Pain points → needs
Needs → requirements

→

03

DESIGN A SOLUTION

Choose where AI helps
Sketch · Prototype

→

04

EVALUATE WITH USERS

Test real tasks
Observe · Improve

Repeat the cycle as you learn. Users should be involved throughout.

Start with a user need — not with a decision to use AI.

Design framework: ISO 9241-210:2019 · Human-centred design for interactive systems

01

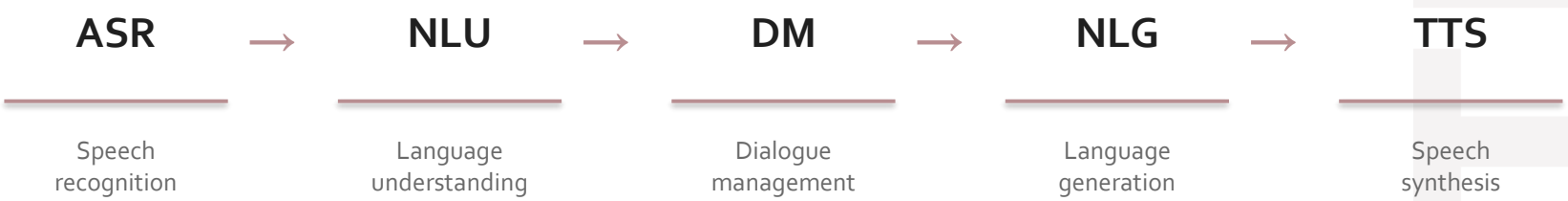
VOICE & SPEECH AI

How AI listens, understands, responds and acts.

A Framework for Spoken Dialogue Systems

Kennington, Lison & Schlangen (2025) · Figure 2

TRADITIONAL MODULAR ARCHITECTURE



Modern end-to-end systems may combine or blur these modules.

Source: Kennington et al. (2025), Dialogue & Discourse, 16(3), 96–130. DOI: 10.5210/dad.2025.305

Google Live Transcribe

ASR · AUTOMATIC SPEECH RECOGNITION

AI CONTRIBUTION

Converts nearby speech and sounds into text in real time.

USER EXPERIENCE

Users read captions while staying in a conversation; they can also type when needed.

ASR · Recognises speech and displays live captions.

ASR · Turns live speech into readable captions.



Amazon Alexa

NLU · NATURAL LANGUAGE UNDERSTANDING

AI CONTRIBUTION

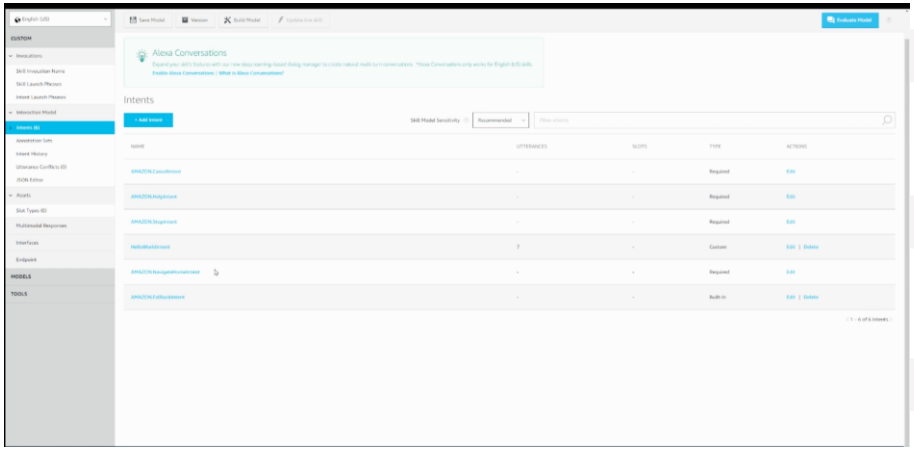
Maps different spoken phrases to an underlying intent and relevant information slots.

USER EXPERIENCE

People can express the same request in different ways instead of memorising one exact command.

NLU (focus) · Maps phrases to intents and slots.

NLU · Identifies intents and information slots.



Voice Agent Builders: Vapi & Retell

DM · BUILDING & MANAGING VOICE AGENTS

AI CONTRIBUTION

Vapi connects speech, language models and tools. Retell offers similar workflows for AI phone agents.

USER EXPERIENCE

Developers define steps, connect APIs and decide when a voice agent should act or hand over to a person.

AI ROLE

DM-enabling platform · Defines dialogue and tool workflows.



Duolingo Video Call with Lily

NLG · NATURAL LANGUAGE GENERATION

AI CONTRIBUTION

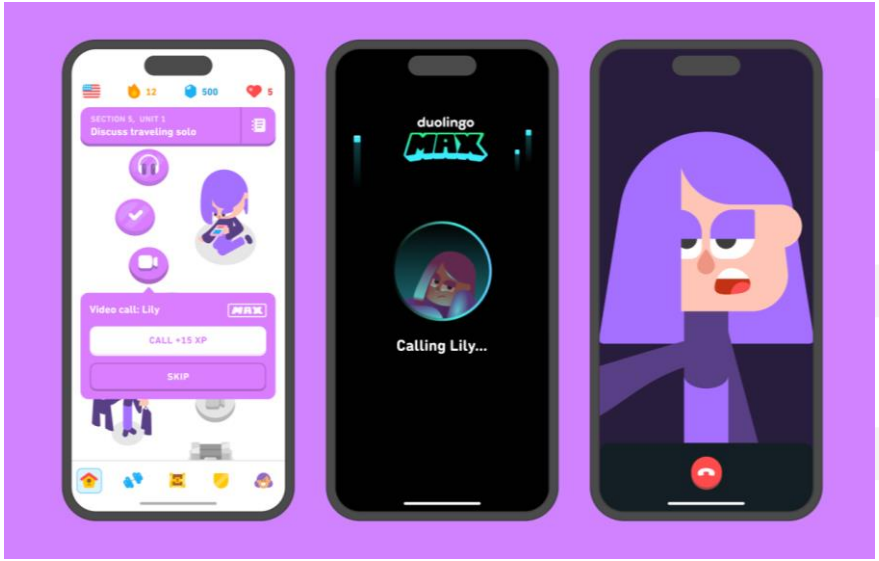
Generates context-appropriate spoken replies for language-learning conversations.

USER EXPERIENCE

Learners practise spontaneous dialogue and can ask Lily to repeat or slow down.

NLG (focus) · Creates replies for speaking practice.

NLG (focus) · Generates language-practice replies.



Speechify: AI Reading Assistance

TTS · TEXT-TO-SPEECH SYNTHESIS

AI CONTRIBUTION

Reads documents and websites aloud with synthetic voices. ElevenLabs shows the same TTS capability for narration.

USER EXPERIENCE

Readers can listen, follow highlighted text and adjust the speed instead of relying on visual reading alone.

TTS (focus) · Reads written information aloud.

TTS · Makes written information available as spoken audio.



Visual Captions: AI Proactivity

RESEARCH EXAMPLE · VOICE AI / HUMAN CONTROL

WHAT IT DOES

Suggests relevant images and visuals during live conversations.

AI ROLE

Speech interpretation + visual suggestions · Beyond the five voice modules.

HUMAN ROLE

Controls when suggested content is shared with others.

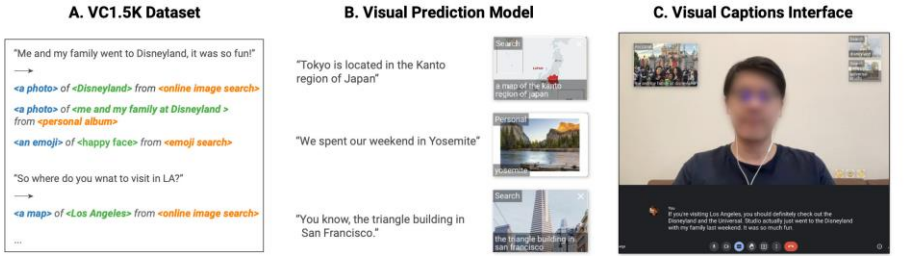


Figure 1: Visual Captions is a real-time system that suggests relevant visuals in conversations. We contribute (A) VC1.5K, a crowdsourced dataset that contains 1595 quadruples of language, visual content, type, and source; (B) a visual prediction model fine-tuned on GPT-3 to suggest relevant visuals, and (C) Visual Captions interface that allows users to share visuals on-the-fly in video conferences.

Liu et al. (2023). Visual Captions. CHI '23. DOI: 10.1145/3544548.3581566

StepWrite: Voice-Guided Writing

RESEARCH EXAMPLE · UIST 2025 · VOICE AI / ADAPTIVE DIALOGUE

WHAT IT DOES

Guides users to write longer text through short spoken questions.

AI ROLE

DM + NLG (using ASR/TTS) · Asks adaptive questions and drafts text.

HUMAN ROLE

Sets the goal, answers questions, and can skip or revise.

HCI VALUE · Lower mental effort



Figure 1: StepWrite enables hands-free, eyes-free writing through an adaptive Q&A dialogue that incrementally elicits task-relevant context and synthesizes coherent drafts. Each row illustrates a three-step interaction: (A) the user performs a hands-busy activity, (B) engages in voice-based Q&A with the AI agent, and (C) receives a personalized draft. The examples shown: (I) walking and (II) cooking highlight two of many multitasking contexts where StepWrite supports hands-free nonvisual composition.

El Alaoui et al. (2025). StepWrite. UIST '25. DOI: 10.1145/3746059.3747610

AROMA: Accessible Cooking with AI

RESEARCH EXAMPLE · UIST 2025 · VOICE AI / ACCESSIBILITY

WHAT IT DOES

Helps blind and low-vision users follow cooking videos.

AI ROLE

Context perception + dialogue/voice · Answers questions and gives alerts.

HUMAN ROLE

Shares sensory cues, asks questions, and chooses actions.

HCI VALUE · Accessible task support

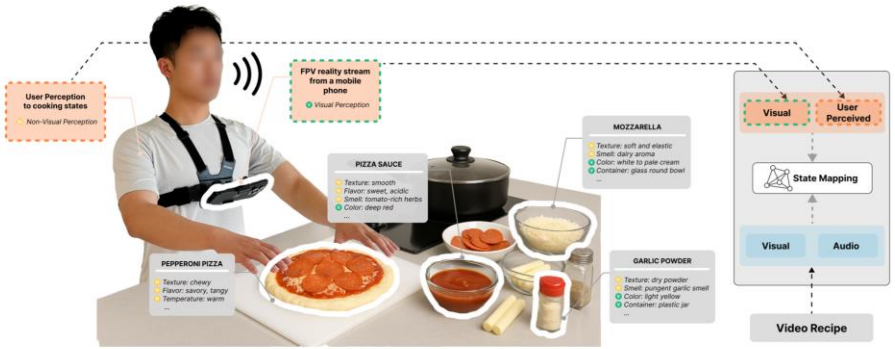


Figure 1: An illustration of how a blind or low-vision (BLV) user uses the AROMA in the kitchen. AROMA helps the user access video recipes by allowing them to communicate their perceived non-visual information (N) about the food, such as texture, smell, taste, etc., along with real-time visual information captured by a wearable camera (V). The system then responds to their questions, referring to the knowledge from the video recipe during cooking. Meanwhile, AROMA proactively monitors the cooking process through the real-time video stream and raises alerts when specific criteria are met.

Ning et al. (2025). AROMA. UIST '25. DOI: 10.1145/3746059.3747650

Voice & Speech AI: Summary

Examples of main functions · Research systems can combine modules and other AI capabilities

MODULE	MAIN FUNCTION	EXAMPLE APPLICATIONS
ASR	Speech → text	Live Transcribe · StepWrite (input)
NLU	Words → meaning / intent	Alexa · Visual Captions (speech cues)
DM	Choose the next response or action	Vapi / Retell · StepWrite · AROMA
NLG	Generate a response	Duolingo Lily · StepWrite · AROMA
TTS	Text → spoken audio	Speechify · StepWrite / AROMA (voice output)

Beyond five modules: visual suggestions (Visual Captions) · context sensing (AROMA).

Voice & Speech AI: Human-Centred Value

What becomes easier, more accessible or more meaningful for users?

HCI BENEFIT	EXAMPLE SYSTEMS	POTENTIAL USER VALUE
Access & independence	Live Transcribe · AROMA	Live captions and context-aware cooking guidance can improve access.
Lower mental effort	StepWrite	Adaptive questions can lower the effort of hands-free writing.
Flexible interaction	Alexa · Visual Captions	Users can express intent naturally or control when AI suggestions appear.
Learning & practice	Duolingo Lily · Speechify	Speaking practice and audio reading support different learning needs.

HCI CHECK: Who benefits? Which task improves? How would you test it?

Mini Exercise 1: Analyse Your Group AI Agent

Use real users and tasks to decide whether voice is useful for your existing Agent.

Refer to P8: ASR · NLU · DM · NLG · TTS

Voice is optional: a text-only Agent may be better for your users.

QUESTION		YOUR NOTES
USER & TASK	Who uses your Agent, and for what task?	
VOICE VALUE	Would speaking make this task easier?	
AI MODULES	Which of ASR, NLU, DM, NLG or TTS apply?	
USER CONTROL	Who corrects output or approves actions?	
YOUR DECISION	Text only or voice enabled? Explain why.	

3–5 minutes · Discuss with your existing project group

02

CREATIVE AI

How AI helps people imagine, generate, evaluate and make.

GenAI Across the Creative Process

Chompunuch & Lubart (2025) · "AI as a Helper"

AI AS A HELPER

The paper identifies four common parts of the creative process by comparing existing models.

Sudapa Chompunuch & Todd Lubart · Journal of Intelligence, 13(5), 57 · 2025

DOI: 10.3390/jintelligence13050057

The framework positions GenAI as support for human creativity — not as an automatic replacement for human judgement.

Four Common Parts of the Creative Process

Classify the AI-supported workflow, not just the type of media it produces

01

PROBLEM IDENTIFICATION & FRAMING

Identify the need or underlying challenge.

→

02

GENERATING IDEAS

Explore concepts, styles and alternatives.

→

03

EVALUATING IDEAS

Compare options, critique and select.

→

04

DEPLOYING & IMPLEMENTING IDEAS

Turn the selected idea into a finished outcome.

Teaching order: generate → evaluate → implement. Problem framing is human-led; stages overlap.

Adapted from Chompunuch & Lubart (2025). J. Intelligence 13(5), 57. DOI: 10.3390/jintelligence13050057

Suno: AI Music Generation

02 GENERATING IDEAS · MUSIC & SOUND

AI CONTRIBUTION

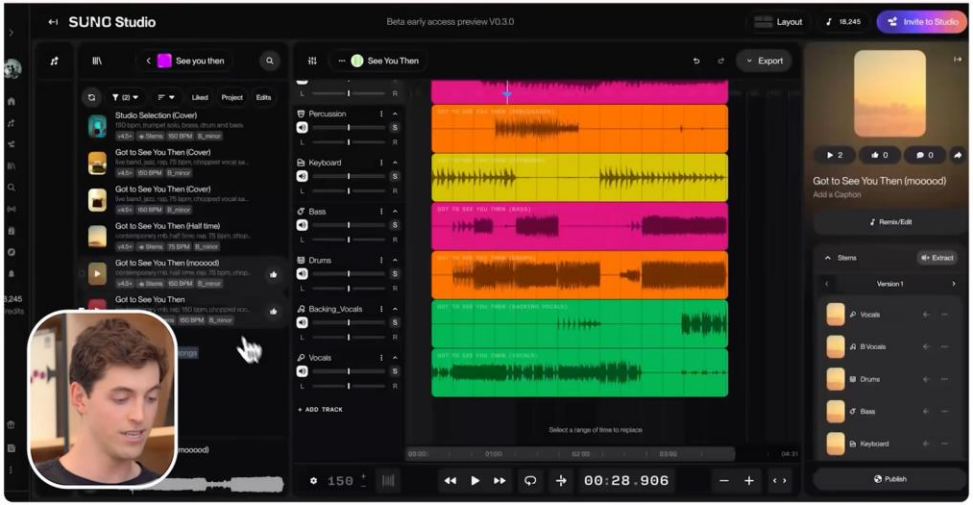
Turns a text idea, mood, or reference into a song with vocals and arrangement; selected sections can be revised.

USER EXPERIENCE

Creators describe the direction, listen to alternatives, and refine parts of the song.

AI ROLE

02 Generating Ideas + 04 Implementing · Song creation.



AI Images: Midjourney & Firefly

02 GENERATING IDEAS · IMAGE CREATION

AI CONTRIBUTION

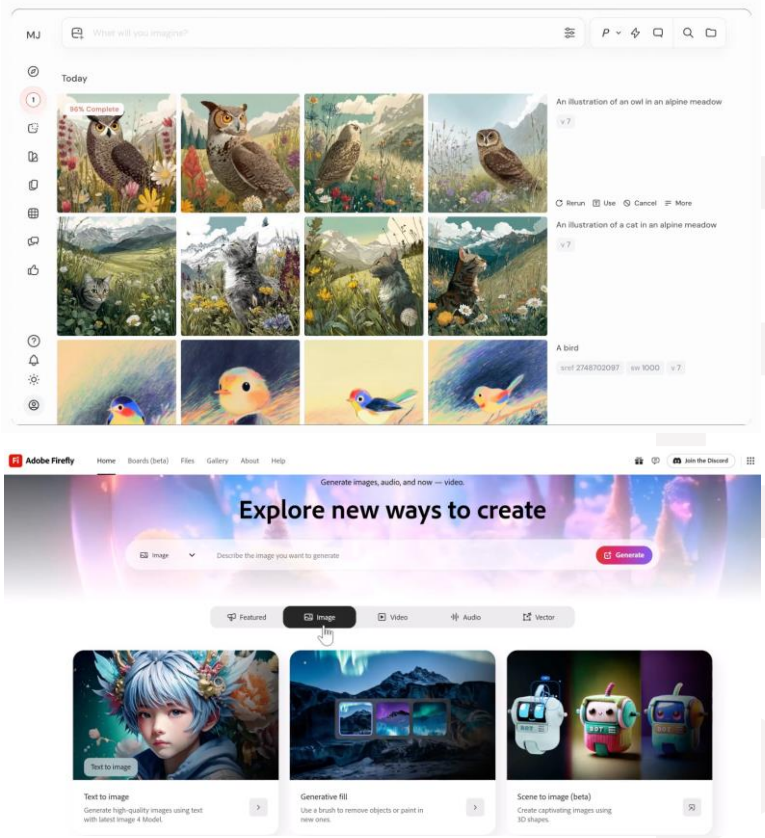
Midjourney generates image ideas from prompts; Adobe Firefly offers image generation and targeted editing.

USER EXPERIENCE

Creators generate alternatives, choose a direction, then refine specific parts when needed.

AI ROLE

02 Generating Ideas + 04 Implementing · Image creation/editing.



Graphic Design: Canva

02 GENERATING IDEAS · GRAPHIC DESIGN

AI CONTRIBUTION

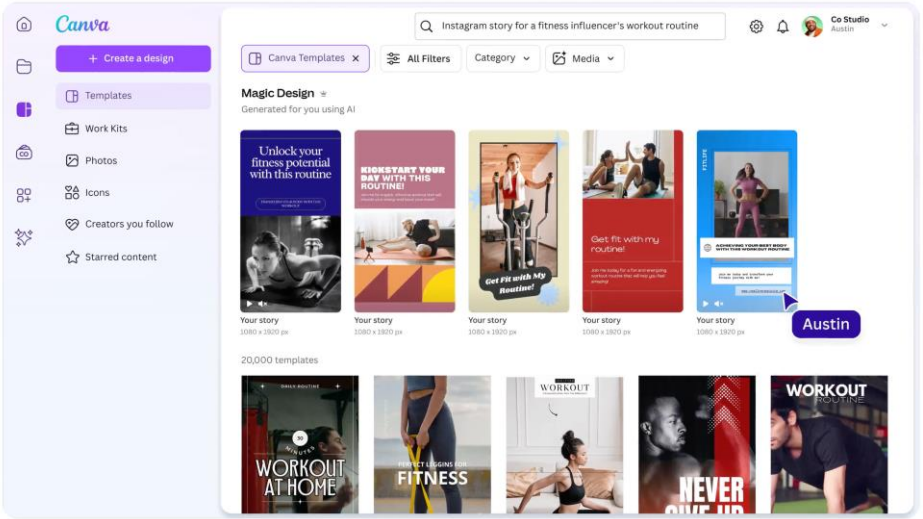
Turn a brief or existing content into design options and help users edit the layout.

USER EXPERIENCE

Beginners can start with templates, change text and images, and prepare a design for sharing.

AI ROLE

02 Generating + 04 Implementing · Drafts and editable designs.



Opal: From Article to Illustration

02 GENERATING IDEAS · ALSO SUPPORTS 04 IMPLEMENTATION

WHAT IT DOES

Helps transform a news article into visual illustrations.

AI ROLE

02 Generating + 04 Partial implementation · Visual concepts and image drafts.

HUMAN ROLE

Explores concepts, selects and edits images before publication.

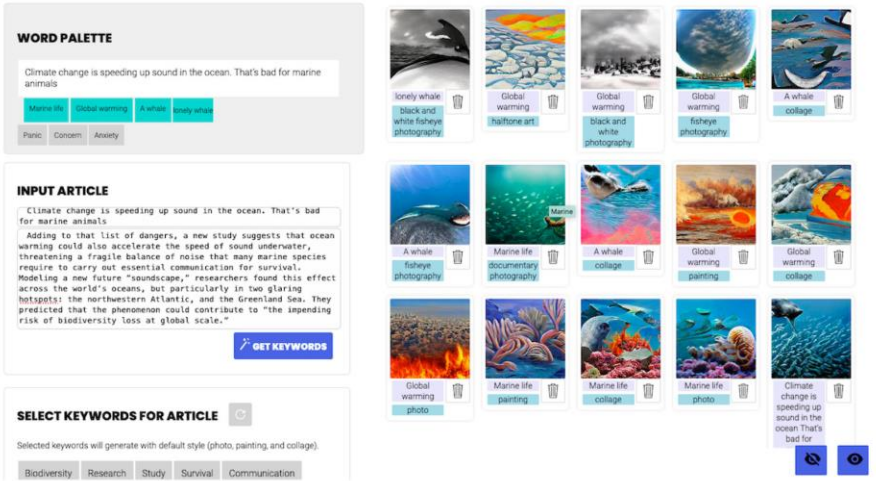


Figure 1: A screenshot of the Opal system, which helps users create news illustrations using a text-to-image generative AI model. The system here has generated a gallery of images for an article on "climate change". The participant is guided through the generation process with a structured pipeline of suggestions based off of GPT-3 generated keywords, tones, and styles.

Liu, Qiao & Chilton (2022). Opal. UIST '22. DOI: 10.1145/3526113.3545621

CreativeConnect: Recombining Ideas

02 GENERATING IDEAS · RESEARCH / REFERENCE RECOMBINATION

WHAT IT DOES

Helps designers recombine visual references into new concepts.

AI ROLE

02 Generating Ideas · Recombines references into new concepts and sketches.

HUMAN ROLE

Selects references, keywords and creative directions.

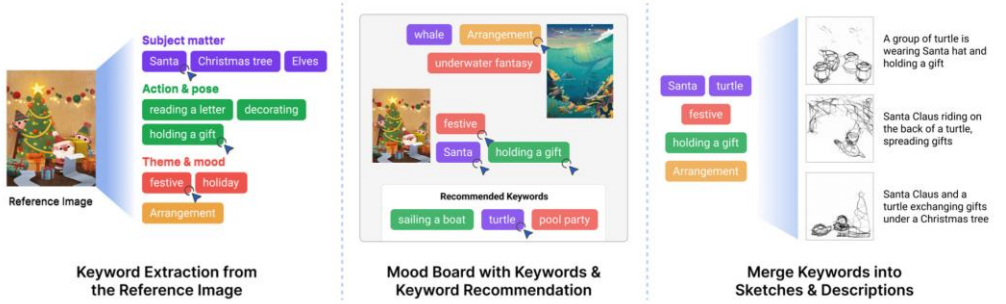


Figure 1: Three main features of CreativeConnect that support reference recombination. (Left) Extraction of 4 different types of keywords from the reference image. (Center) The mood board shows the reference images with the user-selected keywords and keyword recommendations. (Right) Merging selected keywords to generate diverse recombination options and showing them as sketches and descriptions.

Choi et al. (2024). CreativeConnect. CHI '24. DOI: 10.1145/3613904.3642794

PromptMagician: Exploring & Comparing Images

03 EVALUATING IDEAS · RESEARCH / IMAGE COMPARISON

WHAT IT DOES

Helps creators explore and compare AI-generated images.

AI ROLE

02 Generating + 03 Evaluating · Prompt suggestions and guided comparison.

HUMAN ROLE

Sets criteria, compares image results and improves the prompt.

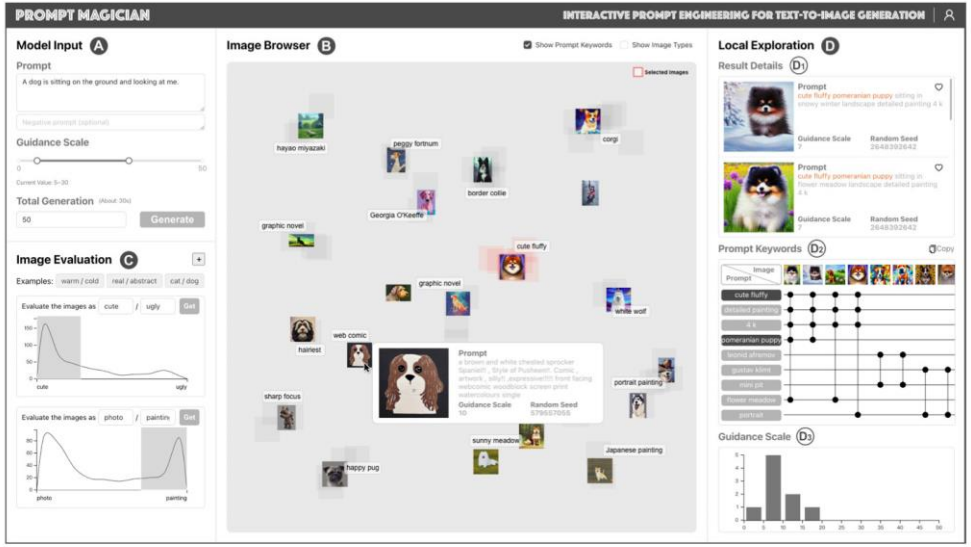


Fig. 1: The user interface of *PromptMagician* consists of four views. The *Model Input View* (A) configures the prompts and hyper-parameters for image creation. The *Image Browser View* (B) visualizes the generated and retrieved images and the recommended prompt keywords. The *Image Evaluation View* (C) helps evaluate and filter images based on multiple criteria. The *Local Exploration View* (D) helps users explore and validate the prompt keywords and guidance scales for images of interest.

Feng et al. (2024). PromptMagician. IEEE TVCG 30(1). DOI: 10.1109/TVCG.2023.3327168

AI Video: Runway & CapCut

04 IMPLEMENTING IDEAS · VIDEO & ANIMATION

AI CONTRIBUTION

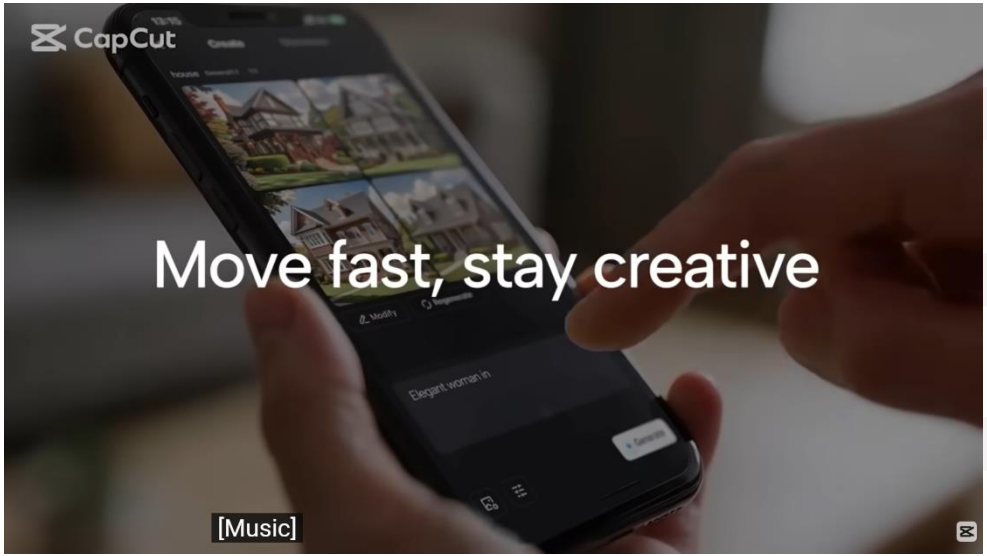
Runway can generate and edit video clips. CapCut adds practical editing tools, including AI captions.

USER EXPERIENCE

Creators make a draft, review the clip, correct problems and export the video.

AI ROLE

04 Implementing (also 02 Generating) · Producing video.



Creative AI: Summary

Teaching order: Generate → Evaluate → Implement · Commercial tools and HCI research

DOMAIN	SYSTEM	TYPE	01 PROBLEM FRAMING	02 GENERATING IDEAS	03 EVALUATING IDEAS	04 IMPLEMENT IDEAS
MUSIC & SOUND	Suno	C	—	●	—	●
VISUAL ART	Midjourney / Firefly	C	—	●	—	●
GRAPHIC DESIGN	Canva	C	—	●	—	●
VISUAL ART	Opal	R	—	●	—	◐
GRAPHIC DESIGN	CreativeConnect	R	—	●	—	—
VISUAL ART	PromptMagician	R	—	●	●	—
VIDEO	Runway / CapCut	C	—	●	—	●

● *Direct system support* ◐ *Partial support* — *Not a core function* *C Commercial* *R Research*

Creative AI: Human-Centred Value

What becomes easier, more accessible or more meaningful for users?

HCI BENEFIT	EXAMPLE SYSTEMS	POTENTIAL USER VALUE
Lower skill barriers	Canva / Adobe Express	AI-generated drafts can help beginners start a design.
Broader exploration	CreativeConnect / Midjourney	Explore and recombine alternatives before choosing a direction.
Judgement & control	PromptMagician / Firefly	Compare results or edit specific parts rather than start over.
Creative expression	Suno / Runway / CapCut	Turn ideas into music or video with less production work.

HCI CHECK: Who benefits? What improves—quality, effort, choice or control?

Mini Exercise 2: Promote Your Group AI Agent

Create ONE PPT slide or poster introducing your existing Agent to potential users.

USE THE FOUR CREATIVE STAGES (P22)

01 FRAME THE PROBLEM

Who is your target user? What need does the Agent address?

02 GENERATE IDEAS

Use AI to explore titles, copy and visual concepts.

03 EVALUATE IDEAS

Choose the clearest message for your audience.

04 IMPLEMENT

Create and edit one slide or poster; check every claim.

ONE-SLIDE PITCH / POSTER

YOUR AGENT NAME

One sentence: why should someone use it?

WHO IT HELPS · PAIN POINT · MAIN BENEFIT

ONE REALISTIC USER REQUEST → AGENT RESPONSE

10–12 minutes • Formative class draft, not a new graded assignment

Peer Feedback: Does Your Agent Pitch Work?

Show your one-slide pitch or poster to another group

1. Who is the target user? Is that clear at first glance?
2. What pain point does the Agent solve?
3. What can the Agent actually do? Is the claim credible?
4. What single change would make the pitch clearer?

Give one specific suggestion, then revise your pitch.

Voice AI vs. Creative AI

Different tasks · Shared questions about human–AI collaboration

DIMENSION	VOICE & SPEECH AI	CREATIVE AI
MAIN GOAL	Communicate, understand, respond and act	Imagine, generate, evaluate and make
AI CONTRIBUTION	Recognise speech, infer meaning, manage turns and generate speech	Generate options, support feedback and make content
HUMAN ROLE	Express intent, clarify, verify and approve	Set direction, select, refine and take responsibility
KEY DESIGN QUESTION	Can users understand and correct the system?	Can creators direct and revise the work?

What Can Go Wrong?

AI outputs can be useful — but they are not automatically correct or appropriate

VOICE & SPEECH AI

- Speech is misrecognised because of noise, accent, or specialist vocabulary.
- The system misunderstands intent or loses conversational context.
- An agent takes the wrong action or fails to hand over.

CREATIVE AI

- Generated content does not match the brief or intended audience.
- Characters, objects, or visual style become inconsistent across edits.
- Outputs may raise originality, bias, attribution, or copyright concerns.

A good AI application makes errors visible and gives people a way to respond.

Designing Generative AI Experiences

Six principles from Weisz et al. (CHI 2024) — what should designers provide?

DESIGN RESPONSIBLY

Whose needs or safety might be affected?

MENTAL MODELS

Can users understand what the AI can do?

APPROPRIATE TRUST

When should users double-check results?

GENERATIVE VARIABILITY

Can people explore different answers?

CO-CREATION

Can users steer and edit the output?

IMPERFECTION

Can users correct errors and recover?

Weisz et al. (2024). Design Principles for Generative AI Applications. CHI '24. DOI: 10.1145/3613904.3642466

Human and AI: Who Should Decide?

Four collaboration modes — choose based on users, tasks and risk (Hu et al., 2025)

AI SUPPORTS

Human makes the final decision.

An Agent suggests an answer.

AI ACTS IF CLEAR

AI handles a bounded task; defers if unsure.

Auto-sort a low-risk request.

AI ASKS FOR INPUT

AI requests needed details before deciding.

Ask which campus or date.

DECIDE TOGETHER

Human and AI work back and forth.

Draft, review and revise.

DESIGNER: set boundaries + recovery USER: express goals + approve important actions

Adapted from Hu et al. (2025), Human at the Center. AI Magazine. DOI: 10.1002/aaai.70043

Trust, Consent & Responsibility

Questions developers should ask before releasing an AI application

01	**DATA & CONSENT**	What is captured, stored, shared, and for how long?
02	**RIGHTS & PROVENANCE**	Do users have permission to use source images, music, voices, and likenesses?
03	**TRANSPARENCY**	Should users or audiences know when content is AI-generated or AI-modified?
04	**ACCOUNTABILITY**	Who reviews, corrects, approves, and takes responsibility for the result?

Check the product's current terms, permissions, and institutional requirements.

Connect Today's Work to Your Group Demo

Refine your existing AI Agent and prepare for its Week 13 live demonstration.

NEXT STEPS FOR YOUR GROUP

1. **Confirm your target users and the core need**
2. **Refine your Agent workflow and boundaries**
3. **Test realistic, tricky and out-of-scope queries**
4. **Improve your Agent introduction and plan a live demo**

Use the pitch slide if helpful; the assessed task remains the live demo and Q&A.

Key Takeaways

Three things to remember when designing AI applications for people

01	**VOICE & SPEECH AI**	AI can recognise speech, understand intent, manage dialogue, generate replies and turn text into speech. One app may combine several roles.
02	**CREATIVE AI**	AI can help generate ideas, compare options and make content. People still define the problem, guide the work and judge the result.
03	**HUMAN-CENTRED DESIGN**	Start with users, tasks and context. Decide what AI does, what people must check or approve, and how users can correct mistakes.

Questions?

This week's tutorial: Bring your group AI Agent or prototype for discussion and peer feedback.